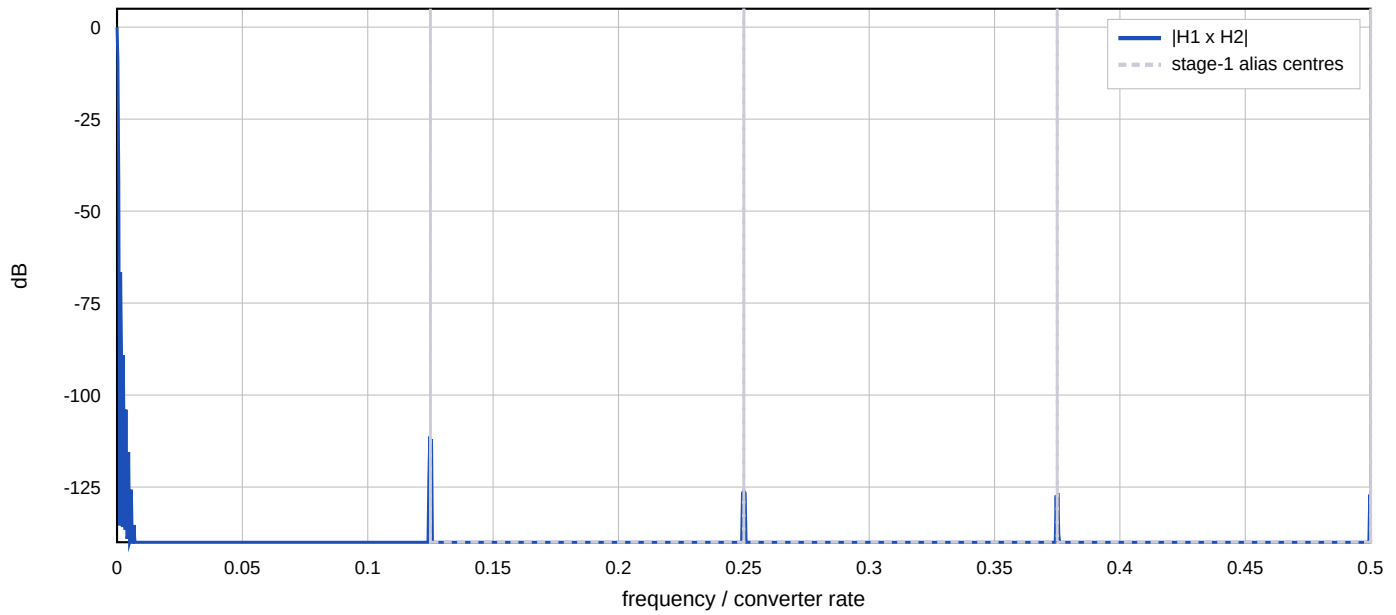


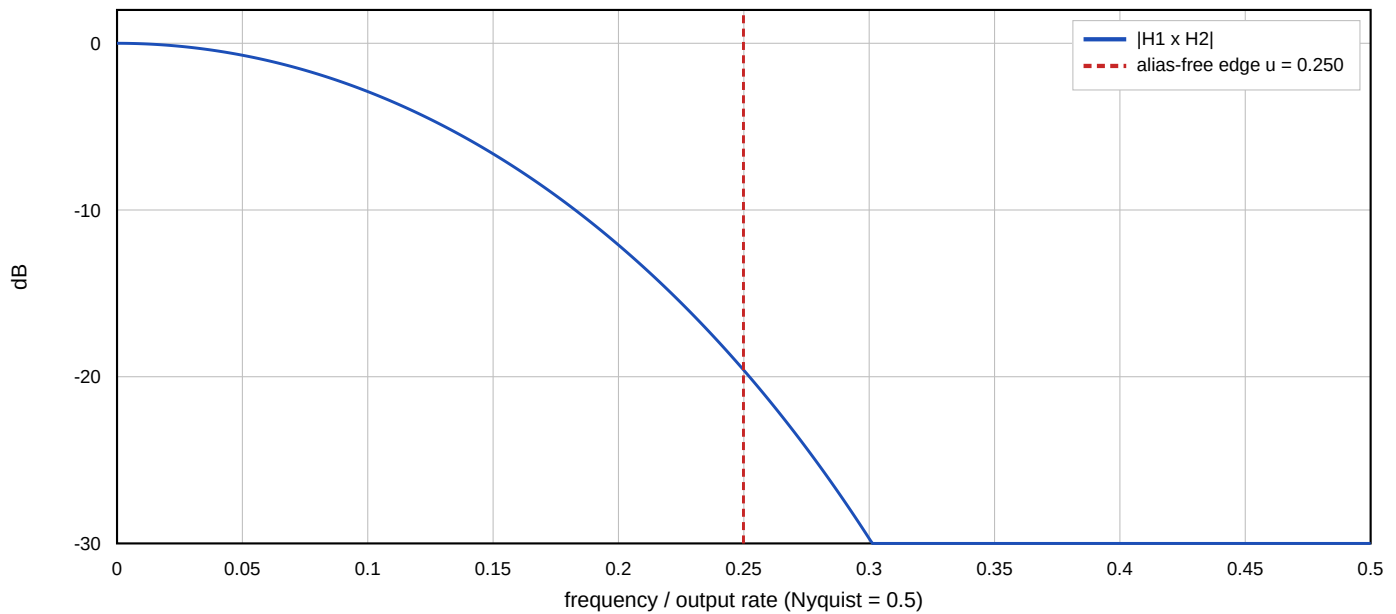
Decimation Chain - Derived Design

125.000 MHz / 488 = 256.1475 kHz, split [8, 61], 64.000 kHz alias-free

Combined magnitude, full input band



Combined magnitude across the decimated band



Stages

stage 1: /8 N=2 M=1 at 125.000 MHz

accumulator 22 bits, prune budget 12, widths [16, 16, 16, 16]

64 register bits, in 16 bits out 16 bits

stage 2: /61 N=5 M=2 at 15.625 MHz

accumulator 51 bits, prune budget 47, widths [37, 30, 24, 18, 16, 16, 16, 16, 16, 16]

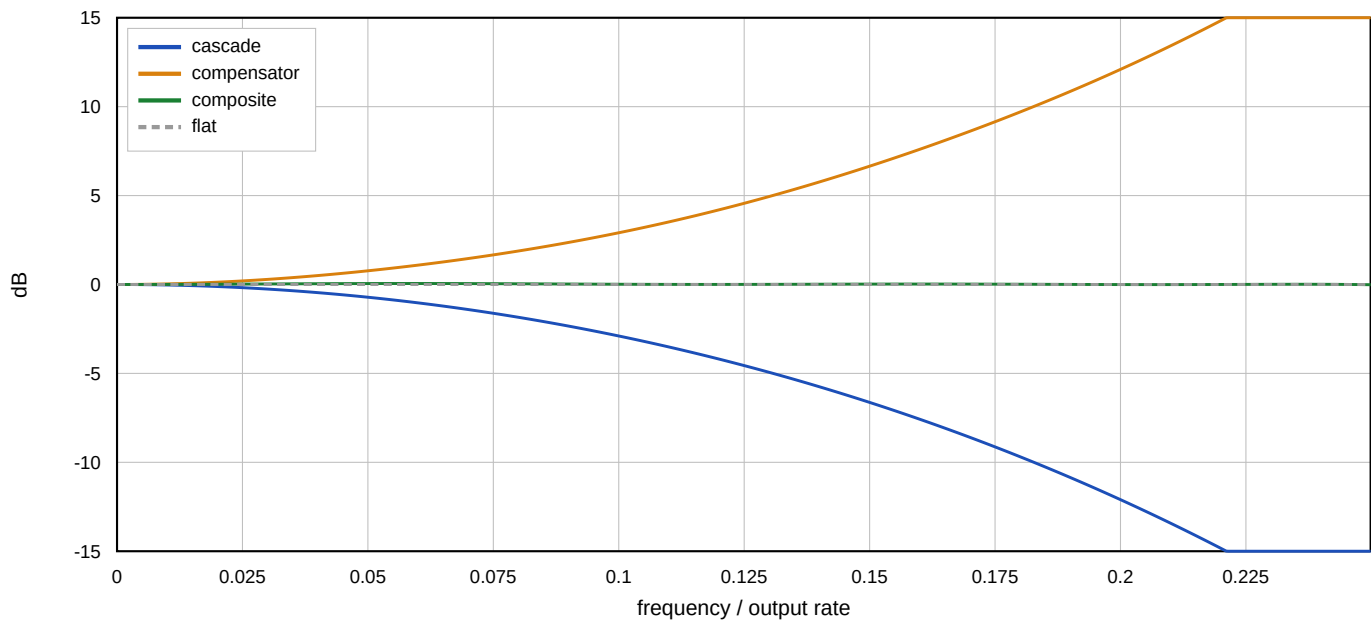
205 register bits, in 16 bits out 16 bits

A stage's nulls must reject the aliases of its own decimation: once energy has folded into the band no later stage can remove it, so every stage carries the full requirement.

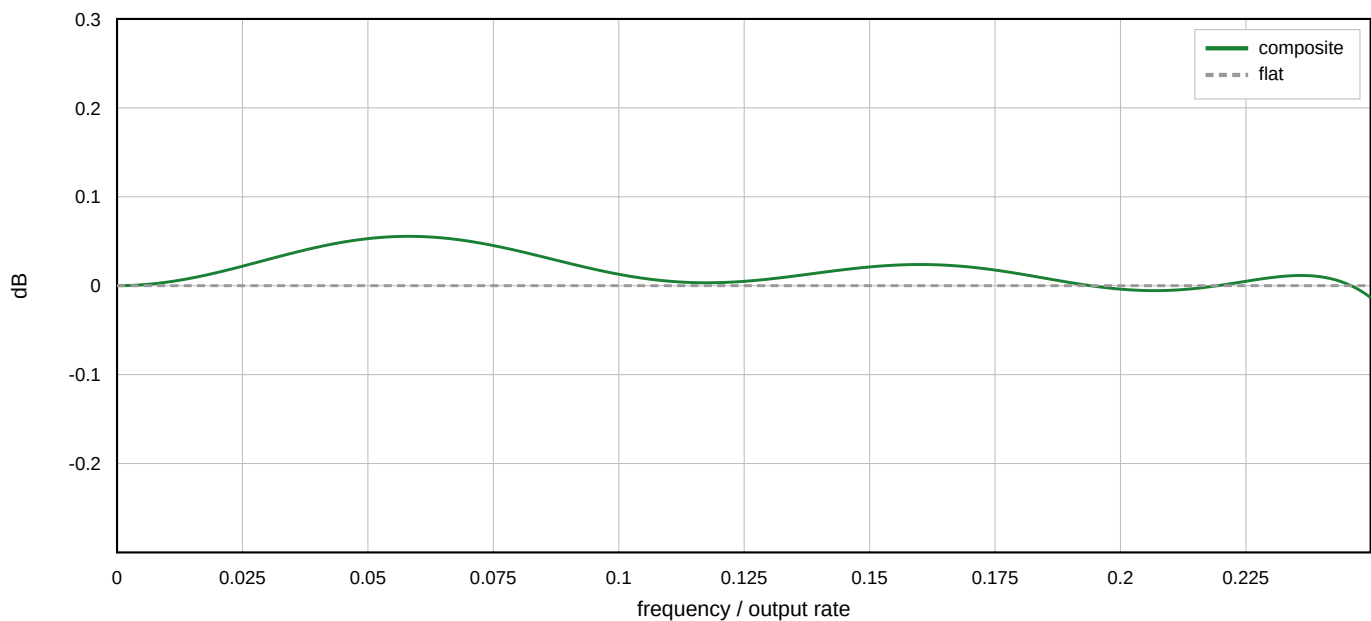
Compensation and Result

11 taps, 16-bit coefficients, 6 multipliers

Cascade, compensator, and the two together



Composite, zoomed - what is left



Achieved

ripple 0.0689 dB (asked ≤ 0.100)

alias rejection ... 67.3 dB (asked ≥ 60.0)

output SNR 84.6 dB (asked ≥ 80.0)

register bits 269 (rate-weighted cost 89.6)

taps [-177, 1286, -4829, 12002, -21186, 26320, -21186, 12002, -4829, 1286, -177]

DC gain 1.000000 (exact by construction)

alternative [4, 61, 2], cost 96.8, 384 bits - feasible, but costlier by the rate-weighted model

Rate-weighted cost is a proxy, not an area figure: flops cost the same however slowly they are clocked. It captures that a wide adder at the converter rate is the hard part.